

CLINTONVILLE, WI, USA

WMO: 726502

Lat: 44.614N

Lon: 88.731W

Elev: 822

StdP: 14.26

Time Zone: -6.00 (NAC)

Period: 97-19

WBAN: 54832

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -8.2	(c) -2.5	(d) -17.3	(e) 2.2	(f) -5.7	(g) -12.9	(h) 2.8	(i) -2.2	(j) 20.8	(k) 18.2	(l) 18.8	(m) 17.2	(n) 4.7	(o) 290	(p) 0.530

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.6	89.6	73.7	85.9	71.0	82.4	69.3	76.1	84.4	74.2	82.3	72.3	80.0	9.5	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
73.2	127.1	80.3	72.1	122.2	79.2	70.1	114.3	77.3	40.2	84.1	38.2	81.9	36.5	79.6	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a) 19.7	(b) 17.3	(c) 15.5	(e) -15.2	(f) 94.2	(g) 4.2	(h) 3.8	(i) -18.3	(j) 97.0	(k) -20.8	(l) 99.2	(m) -23.1	(n) 101.3	(o) -26.2	(p) 104.1
			(e) -15.5	(f) 79.2	(g) 4.2	(h) 3.2	(i) -18.5	(j) 81.5	(k) -21.0	(l) 83.3	(m) -23.3	(n) 85.1	(o) -26.3	(p) 87.4

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	45.7	18.4	21.2	32.3	44.7	56.5	65.9	70.6	68.3	61.3	48.2	35.9	23.5
	DBStd	20.40	11.50	11.08	11.26	9.01	8.52	6.75	5.73	5.57	7.87	8.33	9.19	11.04
	HDD50	3980	979	806	561	205	31	0	0	0	8	137	430	823
	HDD65	7545	1444	1226	1015	610	289	69	16	29	162	525	872	1288
	CDD50	2410	0	0	11	47	232	479	638	568	346	81	8	0
	CDD65	499	0	0	1	2	25	97	189	132	49	4	0	0
	CDH74	4839	0	0	8	43	332	937	1829	1134	511	45	0	0
	CDH80	1413	0	0	0	9	95	266	610	298	129	6	0	0
	WSAvg	6.1	6.5	6.5	6.6	7.3	6.7	5.6	5.1	4.5	5.0	6.2	6.5	6.3
	PrecAvg	31.8	1.3	1.0	1.7	2.8	3.8	4.3	4.1	3.5	3.3	2.7	1.9	1.5
(15) Precipitation	PrecMax	39.0	3.2	1.8	3.4	5.0	6.5	8.3	10.4	9.6	6.7	5.9	4.6	3.1
	PrecMin	27.0	0.2	0.3	0.2	0.6	1.6	1.7	1.1	0.9	0.6	0.2	0.2	0.2
	PrecStd	3.1	0.7	0.4	0.9	1.1	1.3	2.0	2.2	1.7	1.7	1.5	1.1	0.8
	PrecStd	3.1	0.7	0.4	0.9	1.1	1.3	2.0	2.2	1.7	1.7	1.5	1.1	0.8
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.5	51.7	72.3	80.7	88.1	90.4	93.3	90.5	89.6	80.3	66.1	54.0
		MCWB	39.8	45.6	60.6	61.7	69.3	73.3	75.7	74.7	73.4	68.0	56.0	50.9
	2%	DB	39.3	44.6	62.3	72.3	82.0	86.1	89.9	86.4	83.6	72.9	59.4	45.2
		MCWB	35.7	39.6	53.4	57.3	65.8	70.4	74.8	72.1	71.0	62.8	52.7	40.9
	5%	DB	36.3	39.2	54.3	65.9	77.2	82.3	86.2	82.4	80.7	68.0	54.5	40.8
		MCWB	34.2	35.9	47.0	53.3	62.2	69.2	71.7	69.8	67.7	59.8	47.9	37.4
	10%	DB	34.1	36.4	47.7	61.3	72.4	79.5	82.5	80.9	75.5	63.2	49.7	36.8
		MCWB	32.7	33.9	42.0	49.3	60.5	66.5	69.8	68.6	65.5	56.3	45.3	34.7
	0.4%	WB	40.5	46.1	62.1	64.5	72.8	77.0	80.5	77.9	75.2	71.1	58.1	52.1
		MCDB	45.1	49.1	70.1	74.5	82.8	85.7	88.5	85.4	83.7	76.9	63.3	54.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	36.5	40.0	54.4	59.3	68.2	74.3	77.1	74.9	72.6	65.6	53.8	42.0
		MCDB	39.4	43.6	58.8	69.5	77.7	82.6	85.4	82.7	80.5	70.6	58.3	44.0
	5%	WB	34.4	36.9	47.7	54.9	65.6	71.9	74.7	72.8	70.1	60.9	48.7	37.6
		MCDB	35.9	39.5	54.4	63.8	73.7	79.8	82.8	80.2	76.9	66.8	53.7	40.0
	10%	WB	32.5	33.9	42.2	50.8	62.5	69.3	72.5	70.8	67.5	56.7	45.2	35.0
		MCDB	33.7	36.3	47.8	59.9	69.9	76.0	80.1	77.5	74.2	61.8	48.4	36.9
(35) Mean Daily Temperature Range	5% DB	MDBR	14.6	16.3	17.7	20.4	21.6	20.8	21.6	20.7	22.7	18.7	15.2	13.5
		MCDBR	15.5	17.3	25.3	30.7	27.9	24.7	24.4	23.3	26.9	26.5	23.5	16.2
		MCWBR	13.2	13.3	16.3	17.6	13.6	12.1	11.1	11.2	14.0	16.2	16.2	13.4
		MCDBR	14.0	16.0	24.0	27.1	24.1	22.2	21.3	20.4	22.5	22.7	21.1	14.4
	5% WB	MCWBR	12.3	12.7	16.6	17.5	13.2	12.6	11.5	11.4	13.2	16.2	16.1	12.6
		MCWBR	12.3	12.7	16.6	17.5	13.2	12.6	11.5	11.4	13.2	16.2	16.1	12.6
(40) Clear-Sky Solar Irradiance	taub		0.256	0.313	0.330	0.362	0.402	0.401	0.388	0.392	0.366	0.321	0.294	0.265
	taud		2.400	2.215	2.352	2.335	2.267	2.297	2.342	2.327	2.396	2.522	2.512	2.443
	Ebn at Noon		279	274	287	285	276	276	278	272	271	272	263	264
	Edh at Noon		23	34	34	38	42	41	38	38	32	25	21	20
	RadAvg		519	862	1232	1446	1736	1888	1939	1652	1309	825	522	380
(45) All-Sky Solar Radiation	RadStd		56	79	129	148	137	137	94	109	98	84	51	40

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (47 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page